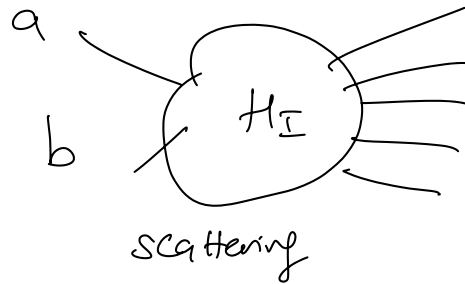
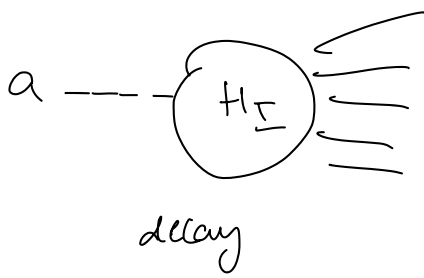


Symmetries



Experimentally: Look at final state $\Rightarrow$ observe some symmetry

Symmetry:

state $|\psi\rangle$

transformation T

$$|\psi'\rangle = T|\psi\rangle$$

$$\langle 0 \rangle = \langle \psi | 0 | \psi \rangle$$

$$\begin{aligned} \langle 0 \rangle &= \langle \psi' | 0 | \psi' \rangle \\ &= \langle \psi | T^\dagger 0 T | \psi \rangle \end{aligned}$$

$[T, H] = 0 \Rightarrow T$ is a symmetry of the system

$$\frac{d}{dt} \langle 0 \rangle = 0$$

Noether's theorem:

a symmetry of the system

$\Leftrightarrow$ a conserved quantity
(constant of motion)

$$|\psi\rangle = |\psi(\underline{q}, \vec{x}, t)\rangle$$

$\hookrightarrow$ quantum numbers

$T|\psi\rangle \approx$ 1) acts on $\vec{x}, t$: external symmetry.
2) act on $\underline{q}$: internal symmetry.

External symmetry:

Rotation in space

$\Leftrightarrow \vec{L}$ angular momentum conserved.

Translation in space

$\Leftrightarrow \vec{p}$

translation in time $\Leftrightarrow E$

space inversion P

(parity)

$$P \psi(\vec{x}, t) = \psi(-\vec{x}, t)$$

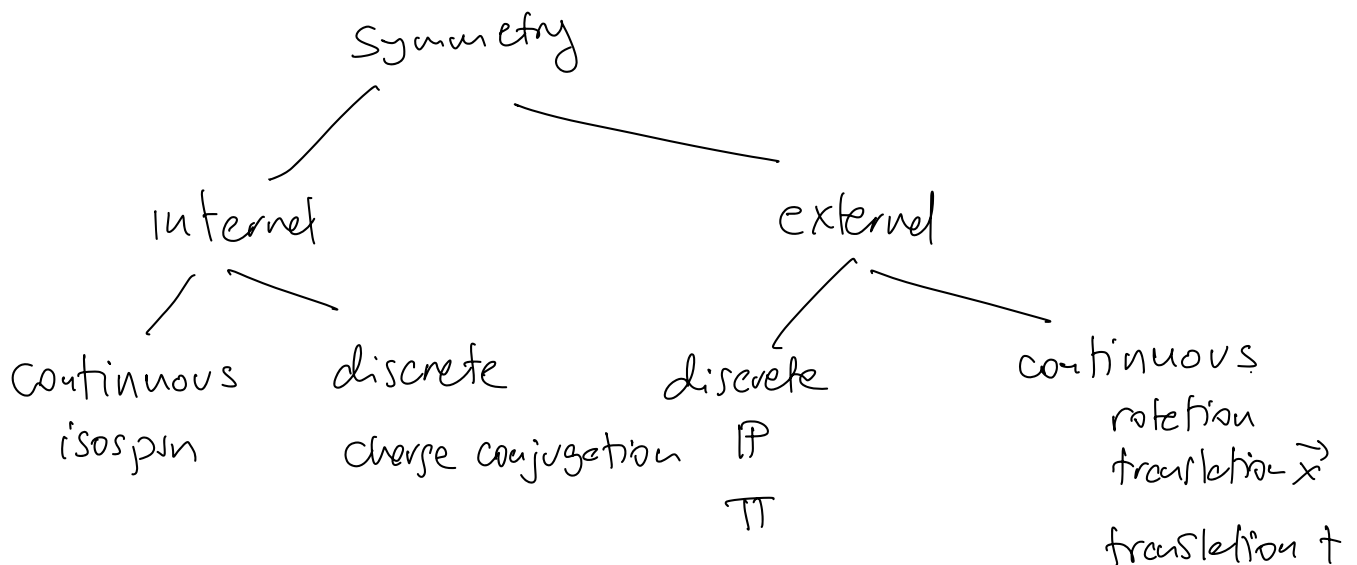
Time reversal T

$$T \psi(\vec{x}, t) = \psi(\vec{x}, -t)$$

Internal symmetry:

internal gauge symm. $\Leftrightarrow$ electric charge

isospin $\Leftrightarrow$



Wigner's theorem:

Symmetry transformations

are $\left. \begin{array}{l} \text{unitary} \\ \text{anti-unitary} \end{array} \right\}$ operators.

contin. transf.

$$T = e^{i \sum a_j G_j} = T(a_1, \dots, a_n)$$

a_j : parameters.
contin.

G_j : generators
of T

$$T^{-1} T = 1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$T: \text{unitary} \Rightarrow \underline{G_j \text{ are Hermitian}} \quad G_j^\dagger = G_j$$

$$\text{space translations} \quad U(\vec{x}) = e^{i \vec{p} \cdot \vec{x}} \quad \vec{p}: \text{generator.}$$

$$\text{rotation} \quad U(\hat{\theta}) = e^{i \hat{\theta} \cdot \vec{L}} \quad \hat{\theta} = (\theta_1, \theta_2, \theta_3)$$

$\vec{L}: \text{angular momentum}$ rotation angles.

$$\text{time translation: } U(t) = e^{i E t} \quad E: \text{generator}$$

contin. operators have additive eigenvalues.

$$U(a) = e^{i a G} \quad \text{one generator.}$$

$$a \rightarrow 0: \quad U(a) = 1 + i a G$$

$$\begin{aligned} U|\psi_1 \psi_2\rangle &= U|\psi_1\rangle U|\psi_2\rangle \\ &= (1 + i a G) |\psi_1\rangle (1 + i a G) |\psi_2\rangle \end{aligned}$$

ψ_1, ψ_2 : are eigenstates

$$G|\psi_1\rangle = g_1 |\psi_1\rangle \quad G|\psi_2\rangle = g_2 |\psi_2\rangle$$

$$= (1 + i a g_1) |\psi_1\rangle (1 + i a g_2) |\psi_2\rangle$$

$$= (1 + i a (g_1 + g_2) + O(a^2)) |\psi_1\rangle |\psi_2\rangle.$$

$$\approx e^{i a (g_1 + g_2)} |\psi_1 \psi_2\rangle$$

$$g_{12} = g_1 + g_2 \quad \text{applies to } \vec{p}, \vec{E}, \vec{L}$$

Discrete operators:

eigenvalues are multiplicative.

$$g_{12} = g_1 \cdot g_2$$

Isospin: internal ant. symmetry

Introduced by Heisenberg in similarity with ang. mom $\vec{J}$
based on experimental observations

1) p, n have similar masses: $m \approx 940 \text{ MeV}$ $\Delta m_{pn} \approx 2 \text{ MeV}$

$$\frac{\Delta m_{np}}{m} \approx 2\%$$

p = uud
n = udd

2) cross section for $\begin{matrix} p + X \\ n + X \end{matrix} \}$ are similar
in strong interaction.

$\Rightarrow$ indication that strong interaction
does not make big distinction between u, p

nucleon = $\begin{pmatrix} p \\ n \end{pmatrix}$ isospin with the same algebra as $\vec{J}$
doublet of isospin

$$I = 1/2$$

$$p = |I=1/2, I_3=+1/2\rangle$$

$$n = |I=1/2, I_3=-1/2\rangle$$

$$|p\rangle = |\uparrow\rangle$$

$$|n\rangle = |\downarrow\rangle$$

$$m_{\pi^+} = m_{\pi^-} \approx 140 \text{ MeV}$$

$$m_{\pi^0} \approx m_{\pi^\pm}$$

$$\frac{\Delta m_{\pi}}{m_{\pi}} = \frac{5 \text{ MeV}}{140 \text{ MeV}} \approx 5\%$$

$$|\pi^+\rangle = |1, +1\rangle$$

$$|\pi^0\rangle = |1, 0\rangle$$

$$|\pi^-\rangle = |1, -1\rangle$$

$$|I, I_3\rangle$$

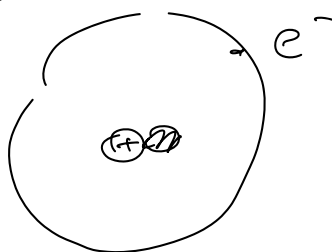
π are a triplet of isospin.

$$\pi \text{ states} = 2I+1$$

Deuterium: ${}^2_1\text{H}$

${}^1_1\text{H}$: hydrogen

deuteron: bound state of $|pn\rangle$
nucleus of deuterium



$$|pn\rangle = |p\rangle |n\rangle$$

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

singlet

triplet.

$$1 = \begin{cases} |++\rangle \\ (|+-\rangle + |-+\rangle) \frac{1}{\sqrt{2}} \\ |--\rangle \end{cases}$$

$$0 = \begin{cases} (|+-\rangle - |-+\rangle) \frac{1}{\sqrt{2}} \end{cases}$$

$$I_3 \text{ of } |pn\rangle = 0$$

$$|pn\rangle = |I, I_3=0\rangle = \alpha |I=0, I_3=0\rangle + \beta |I=1, I_3=0\rangle$$

Suppose $|pn\rangle \equiv (\text{deuteron}) \equiv \text{triplet} = |1, 0\rangle$

$$|pn\rangle = |1, 0\rangle$$

not seen $|pp\rangle = |1, 1\rangle$

not seen $|nn\rangle = |1, -1\rangle$

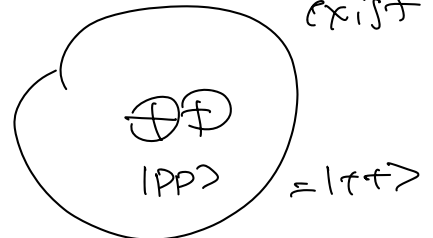
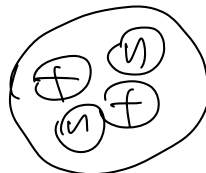
$\Rightarrow$ Indication that $|pn\rangle = |0, 0\rangle$

experim.
 ~~${}^2\text{He}$~~

${}^4_2\text{He}$

experim. does not exist

Helium:



Lecture Tuesday 25/3 @ 8:30

strong interactions conserve I

Hypothesis: $|pn\rangle = |I=0, I_3=0\rangle$

singlet of isospin $I=0$

Experimental test with scattering

$$|d\rangle = |pn\rangle$$

	$p + p \rightarrow d + \pi^+$
Q	$+1 \quad +1 \quad +1 \quad +1$
B	$1 \quad 1 \quad 2 \quad 0$
I_3	$+\frac{1}{2} \quad +\frac{1}{2} \quad 0 \quad +1$

$$\Rightarrow X^{\pm}: \text{non-baryon}, Q \geq \pm 1, I_3 = \pm 1$$

$$\pi^{\pm}$$

$$(a) \quad p + p \rightarrow d + \pi^+$$

$$|i\rangle = |I, I_3\rangle = |1, +1\rangle$$

$$|f\rangle = |d\rangle + |\pi^+\rangle$$

$$0 + 1 = 1$$

$$\langle f | H_I | i \rangle = \langle 1, +1 | H_I | 1, +1 \rangle = \mu$$

$$\sigma(p + p \rightarrow d + \pi^+) \propto |\mu|^2$$

$$I_3(|d\rangle + |\pi^+\rangle) = +1$$

$$I = ? \quad I \neq 0$$

$$I = 1 \quad \checkmark$$

$$|I, I_3 = 0\rangle$$

$$I = 1, 2, 3, 4, \dots$$

(b)

$$\begin{array}{ccc} p + n & \rightarrow & d + X^0 \\ \left. \begin{array}{ccc} Q & 1 & 0 \\ B & 1 & 1 \\ I_3 & +\frac{1}{2} & -\frac{1}{2} \end{array} \right\} & & \left. \begin{array}{ccc} +1 & 0 \\ 2 & 0 \\ 0 & 0 \end{array} \right\} \end{array} \quad X^0 = \pi^0$$

$$|p\rangle + |n\rangle \quad I_3 = 0, \quad \left\{ \begin{array}{l} I = 1 \\ I = 0 \end{array} \right.$$

$$|i\rangle = |p\rangle + |n\rangle = \frac{1}{\sqrt{2}} (|1, 0\rangle + |0, 0\rangle)$$

$$\begin{aligned} |f\rangle = |d\rangle + |\pi^0\rangle &= |I^{(1)} = 0, I_3^{(1)} = 0\rangle + |I^{(2)} = 1, I_3^{(2)} = 0\rangle \\ &= |I^{\text{tot}} = 1, I_3^{\text{tot}} = 0\rangle \end{aligned}$$

$$I^{\text{tot}} = I^{(1)} + I^{(2)}$$

$$\mu_{fi} = \langle f | H_I | i \rangle = \langle 1, 0 | H_I | \frac{1}{\sqrt{2}} (|1, 0\rangle + |0, 0\rangle) \rangle$$

$$\mu_{fi} = \frac{1}{\sqrt{2}} \langle 1, 0 | H_I | 1, 0 \rangle + \frac{1}{\sqrt{2}} \langle 1, 0 | H_I | 0, 0 \rangle$$

if strong interaction
conserves isospin.

$$H_I | 0, 0 \rangle = A | 0, 0 \rangle$$

$$\mu_{fi} = \frac{1}{\sqrt{2}} \langle 1, 0 | H_I | 1, 0 \rangle$$

phase space

$$\frac{\sigma(p+p \rightarrow d+\pi^+)}{\sigma(p+n \rightarrow d+\pi^0)} \sim \frac{|\mathcal{M}(p+p \rightarrow d+\pi^+)|^2}{|\mathcal{M}(p+n \rightarrow d+\pi^0)|^2} \frac{\rho(p+p)}{\rho(p+n)}$$

$$\mathcal{M}(pp) \sim \langle 1, 1 | H_I | 1, 1 \rangle, \quad p+p$$

$$\mathcal{M}(pn) \sim \frac{1}{\sqrt{2}} \langle 1, 0 | H_I | 1, 0 \rangle$$

$$\frac{\rho(pp)}{\rho(pn)}$$

$$p+p \rightarrow d+\pi^+$$

$$p+n \rightarrow d+\pi^0$$

$$\frac{\Delta m_\pi}{m_\pi} \approx 5\%$$

$$\frac{\Delta m_p}{m_p} \approx 2\%$$

$$\frac{\rho(pp)}{\rho(pn)} \approx 1 \text{ within } 5\%$$

$$\Rightarrow \frac{\#(p+p \rightarrow d+\pi^+)}{\#(p+n \rightarrow d+\pi^0)} \approx \frac{1}{|\frac{1}{\sqrt{2}}|^2} \approx 2 \quad \text{confirmed by experiment}$$

$\Rightarrow |d\rangle$ is an isospin singlet

$\Rightarrow$ supports existence of isospin.

H_I : Strong interaction conserving isospin

$$\mu(pp) \sim \langle I=1 | H_I | I=1 \rangle \quad I_3 = +1$$

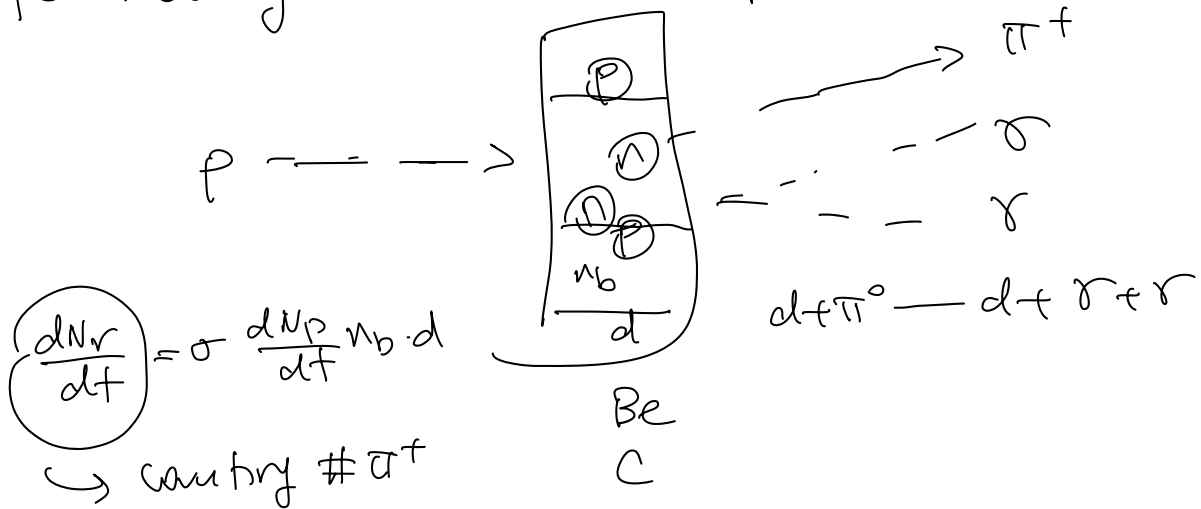
$$\mu(pn) \sim \langle I=1 | H_I | I=1 \rangle \quad I_3 = 0$$

could be different $\langle I=0 | H_I | I=0 \rangle$

Experimentally

$$p+n \rightarrow d+\pi^0$$

$$p+p \rightarrow d+\pi^+$$



$$\#(p+p \rightarrow d+\pi^+) = \sigma(pp) \cdot \Phi_p N_B$$

π -nucleon scattering

π^+ beam against target of p, n

$$\pi^+ + p \rightarrow \pi^+ + p \quad \text{elastic scattering (a)}$$

I	1	$1/2$	1	$1/2$
I_3	+1	+1/2	+1	1/2

$$|I, I_3\rangle \otimes |I, I_3\rangle$$

$$|1/2, 1\rangle |1/2, 1/2\rangle = |3/2, 3/2\rangle$$

$$I_{\text{tot}} = +3/2$$

$$I_3^{\text{tot}} = +3/2$$

~~$$\pi^+ + p \rightarrow \pi^{++} + n$$~~

$$\pi^+ + p \rightarrow n + \pi^+ + \pi^+$$

possible but messy
with CG coeff.

limit $\rightarrow$ 2-body final states

$$\pi^+ + n \rightarrow \pi^+ + n$$

(d)

$$\pi^+ + n \rightarrow \pi^0 + p$$

(g)

Now consider π^-

$$\pi^- + p \rightarrow \pi^- + p$$

(c)

$$\pi^- + p \rightarrow \pi^0 + n$$

(j)

$$\pi^- + n \rightarrow \pi^- + n$$

(f)

~~$$\pi^- + \pi^0$$~~

$\Delta B = 1$